

Day 2- Mockup Exams AI-900

1. Which type of machine learning algorithm finds the optimal way to split a dataset into groups without relying on training and validating label predictions?

- a. Classification
- b. Clustering
- c. Regression

Clustering is a form of *unsupervised* machine learning in which observations are grouped into clusters based on similarities in their data values, or *features*. This kind of machine learning is considered unsupervised because it doesn't make use of previously known *label* values to train a model. In a clustering model, the label is the cluster to which the observation is assigned, based only on its features.

Reference:

<https://learn.microsoft.com/en-us/training/modules/train-evaluate-cluster-models/2-what-is-clustering>

2. Predicting rainfall for a specific geographical location is an example of which type of machine learning?

- a. Classification
- b. Clustering
- c. Regression

Predicting rainfall is an example of regression machine learning, as it will predict a numeric value for future rainfall by using historical time-series rainfall data based on factors, such as seasons. Clustering is a machine learning type that analyzes unlabeled data to find similarities in the data. Classification is used to predict categories of data.

3. A company deploys an online marketing campaign to social media platforms for a new product launch. The company wants to use machine learning to measure the sentiment of users on the Twitter platform who made posts in response to the campaign. Which type of machine learning is this?

- a. Classification
- b. Clustering
- c. Data transformation

Classification is used to predict categories of data. It can predict which category or class an item of data belongs to. In this example, a machine learning model trained by using classification with labeled data can be used to determine the type of bone fracture in a new scan that is not labeled already. Regression is used to predict numeric values. Clustering analyzes unlabeled data to find similarities in the data.

4. A healthcare organization has a dataset consisting of bone fracture scans that are categorized by using predefined fracture types. The organization wants to use machine learning to detect the different types of bone fractures for new scans before the scans are sent to a medical practitioner. Which type of machine learning is this?

- a. Classification
- b. Featurization
- c. Regression

Classification is used to predict categories of data. It can predict which category or class an item of data belongs to. In this example, a machine learning model trained by using classification with labeled data can be used to determine the type of bone fracture in a new scan that is not labeled already. Featurization is not a machine learning type. Regression is used to predict numeric values.

5. A retailer wants to group together online shoppers that have similar attributes to enable its marketing team to create targeted marketing campaigns for new product launches. Which type of machine learning is this?

- a. Classification
- b. Clustering
- c. Multiclass classification

Clustering is a machine learning type that analyzes unlabeled data to find similarities present in the data. It then groups (clusters) similar data together. In this example, the company can group online customers based on attributes that include demographic data and shopping behaviors. The company can then recommend new products to those groups of customers who are most likely to be interested in them. Classification and multiclass classification are used to predict categories of data.

6. In a regression machine learning algorithm, how are features and labels handled in a validation dataset?

- a. Features are compared to the feature values in a training dataset.
- b. Features are used to generate predictions for the label, which is compared to the actual label values.
- c. The label is used to generate predictions for features, which are compared to the actual feature values.

In a regression machine learning algorithm, features are used to generate predictions for the label, which is compared to the actual label value. There is no direct comparison of features or labels between the validation and training datasets.

7. Which assumption of the multiple linear regression model should be satisfied to avoid misleading predictions?

- a. Features are dependent on each other.
- b. Features are independent of each other.
- c. Labels are independent of each other.

Multiple linear regression involves two or more independent variables that contribute to a single dependent variable. Problems in which multiple inputs are used to predict a single numeric outcome are also called *multivariate linear regression*.

Reference:

<https://learn.microsoft.com/en-us/azure/machine-learning/component-reference/linear-regression?view=azureml-api-2>

8. A company wants to predict household water use based on the number of people in a house, the weather temperature, and the time of year. In terms of data labels and features, what is the label in this use case?

- a. Number of people in the house
- b. Time of year
- c. Water use

Water use is the label value that you want to predict, also known as the independent variable. Number of people in the house, weather temperature, and time of year are features, and are values that are dependent on the label. Number of people in the house, weather temperature, and time of year can influence the water consumed in a household.

9. You need to identify numerical values that represent the probability of humans developing diabetes based on age and body fat percentage. Which type of machine learning model should you use?

- a. Linear Regression
- b. Logistic Regression
- c. Multiple Linear Regression

10. Which three values are returned by the language detection feature of the Language service in Azure?

- Bounding box coordinates
- ISO 6391 code
- Language name
- Score

Language Name, ISO 6391 Code, and Score are three values returned by the Language service of natural language processing (NLP) in Azure. Bounding box coordinates are returned by the computer vision services in Azure.

11. Which two features of Azure Cognitive Services allow you to identify issues from support question data, as well as identify any people and products that are mentioned? Each correct answer presents part of the solution.

- a. Azure Bot Service
- b. Conversational Language Understanding
- c. Key Phrase Extraction
- d. Named Entity Recognition

Key phrase extraction is used to extract key phrases to identify the main concepts in a text. It enables a company to identify the main talking points from the support question data and allows them to identify common issues. Named entity recognition can identify and categorize entities in unstructured text, such as people, places, organizations, and quantities. Conversational Language Understanding and Azure Bot Service are not designed for identifying key phrases or entities.

12. You need to use the Azure Machine Learning designer to train a machine learning model. What should you do first in the Machine Learning designer?

- a. Add a dataset
- b. Add a training modules
- c. Create a pipeline

Before you can start training a machine learning model, you must first create a pipeline in the Machine Learning designer. This is followed by adding a dataset, adding training modules, and eventually deploying a service.

13. What should you do after preparing a dataset and before training the machine learning model?

- a. Clean missing data
- b. Normalize the data
- c. Split data into training and validation datasets

Splitting data into training and validation datasets leaves you with two datasets, the first and largest of which is the training dataset you use to train the model. The second, smaller dataset is the held back data and is called the validation dataset, as it is used to evaluate the trained model. If normalizing or summarizing the data is required, it will be carried out as part of data transformation. Cleaning missing data is part of preparing the data and the data transformation processes.

14. You need to use the Azure Machine Learning designer to deploy a predictive service from a newly trained model. What should you do first in the Machine Learning designer?

- a. Add a dataset
- b. Add training modules
- c. Create an inference pipeline

To deploy a predictive service from a newly trained model by using the Machine Learning designer, you must first create a pipeline in the Machine Learning designer. Adding training modules by using the Machine Learning designer takes place before creating a trained model, which already exists. Adding a dataset by using the Machine Learning designer requires that a pipeline already exists.

15. What is an unsupervised machine learning algorithm module for training models in the Azure Machine Learning designer?

- a. Classification
- b. K-means clustering
- c. Linear regression

K-means clustering is an unsupervised machine learning algorithm component used for training clustering models. You can use unlabeled data with this algorithm. Linear regression and classification are supervised machine learning algorithm components. You need labeled data to use these algorithms.

16. Your company is exploring the use of voice recognition technologies in its smart home devices. The company wants to identify any barriers that might unintentionally leave out specific user groups. This an example of which Microsoft guiding principle for responsible AI?

- a. accountability
- b. Fairness
- c. Inclusiveness
- d. privacy and security

C - Inclusiveness. No one is left out (disabled, gender, ethnicity, LGBTQIA+ etc etc)

Reference:

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

17. What are three Microsoft guiding principles for responsible AI? Each correct answer presents a complete solution.

NOTE: Each correct selection is worth one point.

- a. Knowledgeability
- b. Decisiveness
- c. Inclusiveness
- d. Fairness
- e. Opinionatedness
- f. reliability and safety

The six guiding principles are:

1. Fairness
2. Inclusiveness
3. Transparency
4. Privacy and Security
5. Reliability and Safety
6. Accountability

Reference:

<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/4-guiding-principles>

18. When you design an AI system to assess whether loans should be approved, the factors used to make the decision should be explainable. This is an example of which Microsoft guiding principle for responsible AI?

- a. Transparency
- b. Inclusiveness
- c. Fairness
- d. privacy and security

Achieving transparency helps the team to understand the data and algorithms used to train the model, what transformation logic was applied to the data, the final model generated, and its associated assets. This information offers insights about how the model was created, which allows it to be reproduced in a transparent way.

Incorrect Answers:

B: Inclusiveness mandates that AI should consider all human races and experiences, and inclusive design practices can help developers to understand and address potential barriers that could unintentionally exclude people. Where possible, speech-to-text, text-to-speech, and visual recognition technology should be used to empower people with hearing, visual, and other impairments.

C: Fairness is a core ethical principle that all humans aim to understand and apply. This principle is even more important when AI systems are being developed. Key checks and balances need to make sure that the system's decisions don't discriminate or run a gender, race, sexual orientation, or religion bias toward a group or individual.

D: A data holder is obligated to protect the data in an AI system, and privacy and security are an

integral part of this system. Personal needs to be secured, and it should be accessed in a way that doesn't compromise an individual's privacy.

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai> <https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/strategy/responsible-ai>

19. You are building an AI-based app. You need to ensure that the app uses the principles for responsible AI. Which two principles should you follow? Each correct answer presents part of the solution.

NOTE: Each correct selection is worth one point.

- a. Implement an Agile software development methodology
- b. Implement a process of AI model validation as part of the software review process
- c. Establish a risk governance committee that includes members of the legal team, members of the risk management team, and a privacy officer
- d. Prevent the disclosure of the use of AI-based algorithms for automated decision making

B ensures reliability and safety principle and C ensures privacy and security principle of AI.

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>
<https://docs.microsoft.com/en-us/learn/modules/responsible-ai-principles/3-implications-responsible-ai-practical>

20. Select the answer that correctly completes the sentence.

According to Microsoft's _____ principle of responsible AI, AI systems should NOT reflect biases from the data sets that are used to train the systems.

- a. Accountability
- b. Fairness
- c. Inclusiveness
- d. Transparency

Reference:

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/innovate/best-practices/trusted-ai>